

Test 9 - Answers & Mnemonics

Questions 13, 14, and 60 intentionally omitted.

Source note: Question 19 is listed as OPEN LOOP in the study guide, which also notes a handwritten question about that answer.

Actual Answers

1. HOLDING torque
2. DETENT torque
3. OUT OF PHASE WITH the voltage; LAGGING power factor
4. SERVO MOTORS
5. BUS topology
6. RING topology
7. CONTROL-NET
8. STAR topology
9. 5, 6, and 7
10. 3
11. Between DECIDE WHAT TYPE OF TEST TO PERFORM and VERIFY CORRECT OPERATION
12. 4 LEVELS OF TESTING
15. 50-100 ft apart
16. Up to 18 in. above the floor, extending 20 ft horizontally in all directions
17. Weight of a volume of vapor/gas compared with an equal volume of dry air at the same temperature and pressure
18. A portion of the security system operating under a different program than the rest
19. OPEN LOOP
20. OPEN
21. SUPERVISORY SYSTEM
22. SPOT PROTECTION
23. SPECIFIC AREA PROTECTION
24. POINT OF ENTRY
25. MIXED CREDENTIAL technology
26. A COMPUTER
27. AUTHENTICATION
28. MODEM
29. DTMF signal
30. MULTIPLEX DVR
31. DISPLAY RESOLUTION
32. Allows the emergency control panel to take control of the telephone line
33. PAN-TILT-ZOOM (PTZ) camera

- 34. CENTRAL AMPLIFIED SYSTEM**
- 35. RESISTIVE TOUCH-SCREEN CRT**
- 36. GEARED TRACTION elevator**
- 37. GOVERNOR**
- 38. PHASE TWO MODE**
- 39. INDEPENDENT SERVICE MODE**
- 40. CONVENIENCE, EFFICIENCY, SAFETY, and SERVICE**
- 41. ELEVATOR SHUT DOWN / SHUNT TRIP**
- 42. LOSS OF POWER and LOW VOLTAGE**
- 43. VOLTAGE DROP OF MORE THAN 5%**
- 44. TURN POWER OFF, LOTO**
- 45. TROUBLESHOOTING**
- 46. VIEW THE PROGRAM**
- 47. APPEARS EMPTY**
- 48. F1**
- 49. AT THE BEGINNING OF A PROGRAM**
- 50. DIALOGUE BOXES**
- 51. PROGRAM MODE**
- 52. Some PLC applications must be stopped in a SPECIFIC SEQUENCE**
- 53. POWER and RUN LEDs**
- 54. ANNUALLY**
- 55. PLC enclosure, PLC-related components, INPUT DEVICES, and OUTPUT COMPONENTS**
- 56. RUN MODE = operating cycle runs continuously to control the machine/process**
- 57. SOFTWARE AND PROGRAM VERIFICATION = ensures the PLC can be accessed by a PC and the PC has a current copy of the PLC program**
- 58. Information about a specific instruction associated with a specific address**
- 59. Localization of an overcurrent condition to restrict outages to the affected circuit/equipment through proper OCPD selection, ratings, and settings**
- 61. LIFE SAFETY and PUBLIC SAFETY**
- 62. More than 150 V to ground, but not exceeding 1,000 V phase-to-phase**
- 63. FIRE PUMP CIRCUITS; also not required for a continuous industrial process where non-orderly shutdown increases hazards**
- 64. A continuous industrial process where non-orderly shutdown would introduce additional or increased hazards**
- 65. ZERO-SEQUENCE and RESIDUAL methods**

Mnemonics

1. HOLD = energized holds position.
2. DETENT = Dead/de-energized torque.
3. Capacitive? 'Out & Lag' - use the study-guide pairing: out of phase + lagging.
4. SERVO Sees position - servo needs feedback.
5. BUS = one backbone bus route.
6. RING = round closed loop.
7. CONTROL-NET = everyone controls/accesses the same source.
8. STAR = everything points to the center.
9. Diverge = 5-6-7 spread out.
10. Feedback = 3 comes back to me.
11. Test -> Correct -> Verify.
12. Funnel = FOUR levels.
15. Pressure piling: '50 to 100 seals the run.'
16. Garage rule = 18 UP, 20 OUT.
17. Vapor density = Vapor vs. Air weight.
18. PARTition = PART has its own program.
19. OPEN LOOP = series - memorize the study-guide pairing.
20. Open loop = normally OPEN.
21. SUPERVISORY = supervises high-security sites.
22. SPOT = one object/spot.
23. SPECIFIC AREA = one specific location.
24. POINT OF ENTRY = doors/entry points.
25. MIXED = mix two credentials.
26. Database = COMPUTER.
27. AUTHENTICATION = prove who you are.
28. MODEM = MODulate/DEModulate.
29. DTMF = Dial Tones Move Functions.
30. MULTIPLEX = MULTI cameras.
31. Pixels = RESOLUTION.
32. Line seizure = emergency panel SEIZES the line.
33. PTZ = Pan, Tilt, Zoom.
34. Paging starts CENTRAL, then AMPLIFIED.
35. CRT = the odd old monitor out.
36. GEARED = worm GEAR.
37. GOVERNOR governs overspeed/brakes.
38. Phase ONE recall -> Phase TWO manual control.

- 39. INDEPENDENT = independent of central control.
- 40. CESS = Convenience, Efficiency, Safety, Service.
- 41. Water coming? SHUNT it DOWN.
- 42. PLC power problems = LOST or LOW.
- 43. Overloaded transformer = OVER 5% drop.
- 44. Before hardware: OFF + LOTO.
- 45. Forcing = TROUBLESHOOTING.
- 46. First PLC troubleshooting move: VIEW it.
- 47. Forced input = EMPTY box.
- 48. F1 = First key for help.
- 49. Debugging begins at the BEGINNING.
- 50. Choices in windows = DIALOGUE BOXES.
- 51. PROGRAM mode = program ready, process paused.
- 52. Second tech = SEQUENCE support.
- 53. Powered and operating = POWER + RUN.
- 54. PLC maintenance = Annual appointment.
- 55. Visual = E-R-I-O: Enclosure, Related components, Inputs, Outputs.
- 56. RUN = Running continuously.
- 57. Verify = PC ACCESS + CURRENT COPY.
- 58. Instruction comment = instruction + address information.
- 59. Selective coordination = nearest device isolates the problem.
- 61. Selective coordination saves L&P: Life + Public safety.
- 62. GFPE voltage rule: 'Over 150, under/equal 1000.'
- 63. FIRE PUMP + hazardous shutdown = exception.
- 64. Process hazard? Don't force the shutdown.
- 65. Z-R = Zero-sequence + Residual.